

Results

01 · LOGISTIC REGRESSION

predict a yes/no outcome

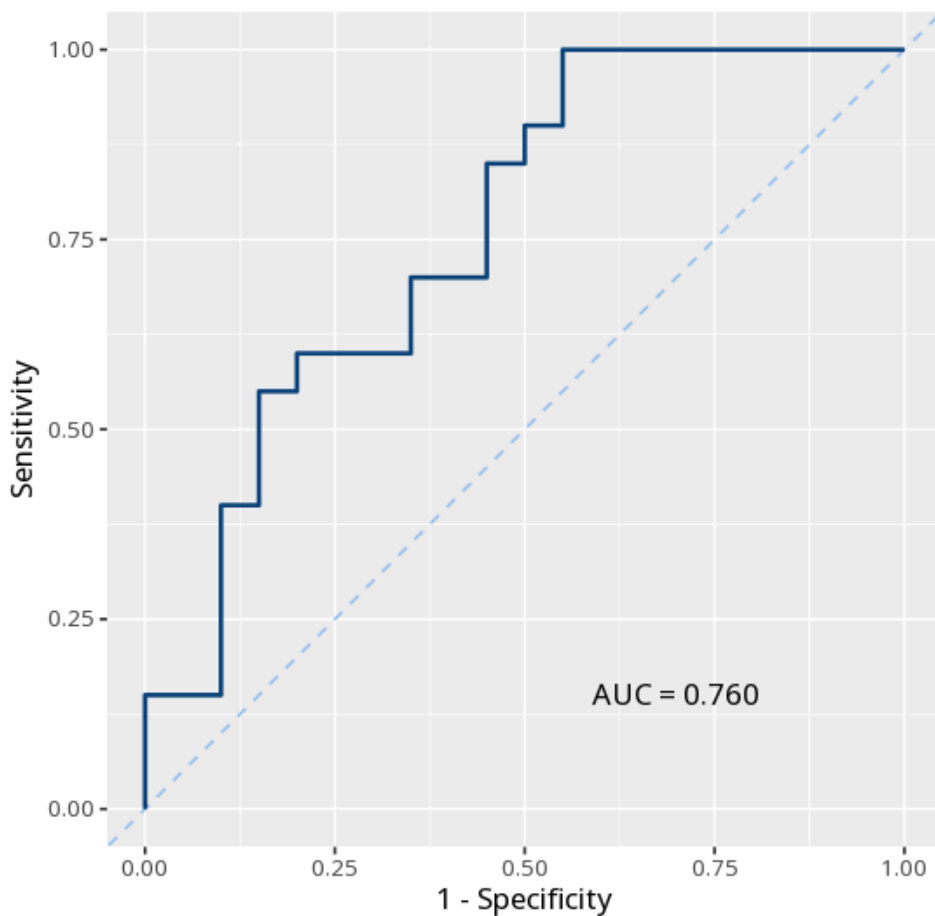
Table 1. Coefficients

	Log-odds (B)	Odds ratio (OR)
(Intercept)	−6.59 (2.92)	< 0.01 [< 0.01, 0.24]
pre_score	0.08 (0.04)	1.08 [1.01, 1.18]
age	0.03 (0.03)	1.03 [0.98, 1.09]
group: b	1.24 (0.72)	3.46 [0.87, 15.44]
Num.Obs.	40	
Nagelkerke R ²	0.28	
Omnibus χ ²	9.54 (p .023)	
Log.Lik.	−22.95	
AIC	53.91	
BIC	60.66	
Accuracy	65.0%	
Sensitivity	65.0%	
Specificity	65.0%	
AUC [95% CI]	.76 [.61, .91]	

Table 2. Classification

Predicted \ Observed	no	yes	% correct
no	13	7	65.0%
yes	7	13	65.0%

Figure. Classifier performance



Understanding the numbers

Odds ratio (OR). How the odds of the outcome multiply per one-unit increase in the predictor (>1 raises the odds, <1 lowers them). Here, OR = 1.08, so a one-unit increase in the predictor multiplies the odds by this factor.

Log-odds (B). The raw coefficient on the log-odds scale that the odds ratio is exponentiated from. Here, B = 0.08 on the log-odds scale.

Omnibus χ^2 . Tests the model as a whole against a model with no predictors. Here, the omnibus $\chi^2 = 9.54$ (p .023).

Nagelkerke R². A pseudo-R² approximating the share of variance explained (0 to 1, higher is more). Here, Nagelkerke R² = 0.28.

Log.Lik.. The log-likelihood: how probable the observed outcomes are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = -22.95.

AIC. A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 53.91.

BIC. Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here, BIC = 60.66.

N. The number of observations used to fit the model. Here, N = 40.

Predicted \ Observed (first category). The diagonal (correct) cell for the first outcome category: how many cases the model predicted into it that were actually observed there. Here, 13 cases were predicted and observed in the first category.

Predicted \ Observed (second category). The diagonal (correct) cell for the second outcome category: how many cases the model predicted into it that were actually observed there. Here, 13 cases were predicted and observed in the second category.

% correct. The overall share of cases the model classified correctly - model fit and the ROC/AUC show how well it classifies overall. Here, the model classified 65.0% of cases correctly.

Accuracy. The overall share of cases classified correctly (same number as % correct, reported as a named scalar here). Here, accuracy = 65.0%.

Sensitivity. The share of ACTUAL events the model correctly predicted as events (the true positive rate). Here, sensitivity = 65.0%.

Specificity. The share of ACTUAL non-events the model correctly predicted as non-events (the true negative rate). Here, specificity = 65.0%.

AUC [95% CI]. The area under the ROC curve, with its confidence interval (pROC's DeLong method) - 0.5 is chance, 1.0 is perfect discrimination. Here, AUC = .76 [.61, .91].

How to read this test

Each predictor's odds ratio tells how the odds of the outcome change per unit (>1 increases, <1 decreases); a term carries weight when its OR confidence interval excludes 1, and the inline omnibus χ^2 (with its p) tests the model as a whole. Model fit and the ROC/AUC show how well it classifies overall, alongside accuracy (overall % correct), sensitivity (% of actual events correctly predicted), and specificity (% of actual non-events correctly predicted). Your significance threshold (α) is 0.05.

APA template: Predictor pre_score was associated with the outcome, OR=1.08, 95% CI [1.01, 1.18], p = .035 (AUC=.76).

Statistical basis: Logistic regression coefficients (log-odds / odds ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Nagelkerke pseudo- R^2 (Lüdtke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.); ROC curve / AUC (Robin X, Turck N, Hainard A, et al. (2011). "pROC: an open-source package for R and S+ to analyze and compare ROC curves." BMC Bioinformatics, 12, 77.). Full references in the exported CITATIONS.txt.

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